No-GPS Drone Project

Technical Report

GPS-Denied Autonomous UAV Localisation & Object Detection

Location: Chiayi, Taiwan 23.4509°N, 120.2861°E

Stack: Isaac Sim 6.0 · AnyLoc · YOLOv8 · ArduPilot MAVLink

Competition: 2nd UAV Defense Challenge (ROC MND) — GPS-denied multi-target detection

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Validated in NVIDIA Isaac Sim before real-hardware deployment

Contents

Project Overview

This project builds a drone system that can **localise itself and detect objects without GPS**, using visual place recognition (AnyLoc), object detection (YOLOv8), and ArduPilot for flight control. The full pipeline is validated in NVIDIA Isaac Sim before deployment to real hardware.

Motivation — Competition Context

The system is developed for the *2nd UAV Defense Application Challenge* (2nd National Defense UAV Challenge, ROC) organised by the Republic of China Ministry of National Defense. The competition theme is *GPS-denied autonomous UAV multi-target reconnaissance*: a GNSS jammer is physically mounted on the drone, and teams must still autonomously navigate, identify, and geo-locate targets.

Table 1: Competition requirements vs. project status

Requirement	Project approach	Status
GPS-denied navigation	AnyLoc + Visual Odometry	Done
Autonomous flight	ArduPilot EKF3 + GUIDED mode	In progress
Vehicle detection (cars)	YOLOv8l (VisDrone)	Done
Target geo-location	AnyLoc anchor + nadir projection	Done
Localisation accuracy ($\leq$50 m)	15–20 m anchor error	Meets requirement

Software Stack

Isaac Sim NVIDIA Isaac Sim 6.0.0 (Kit 106, Python 3.12) — physics simulation, Cesium terrain, nadir camera output.

AnyLoc Universal visual place recognition — DINOv2 ViT-B/14 + VLAD + FAISS — provides GPS-free localisation at 15–20 m accuracy.

YOLOv8 `yolov8l_visdrone.pt` — YOLOv8-Large pre-trained on VisDrone 2019 DET (10 aerial vehicle classes).

ArduPilot Open-source autopilot — SITL JSON backend connects to Isaac Sim; EKF3 fuses AnyLoc position via `VISION_POSITION_ESTIMATE`.

pymavlink MAVLink protocol library — connects Python control code directly to ArduPilot over TCP.

System Pipeline

Figure 1 shows the full data flow from simulation to flight commands.

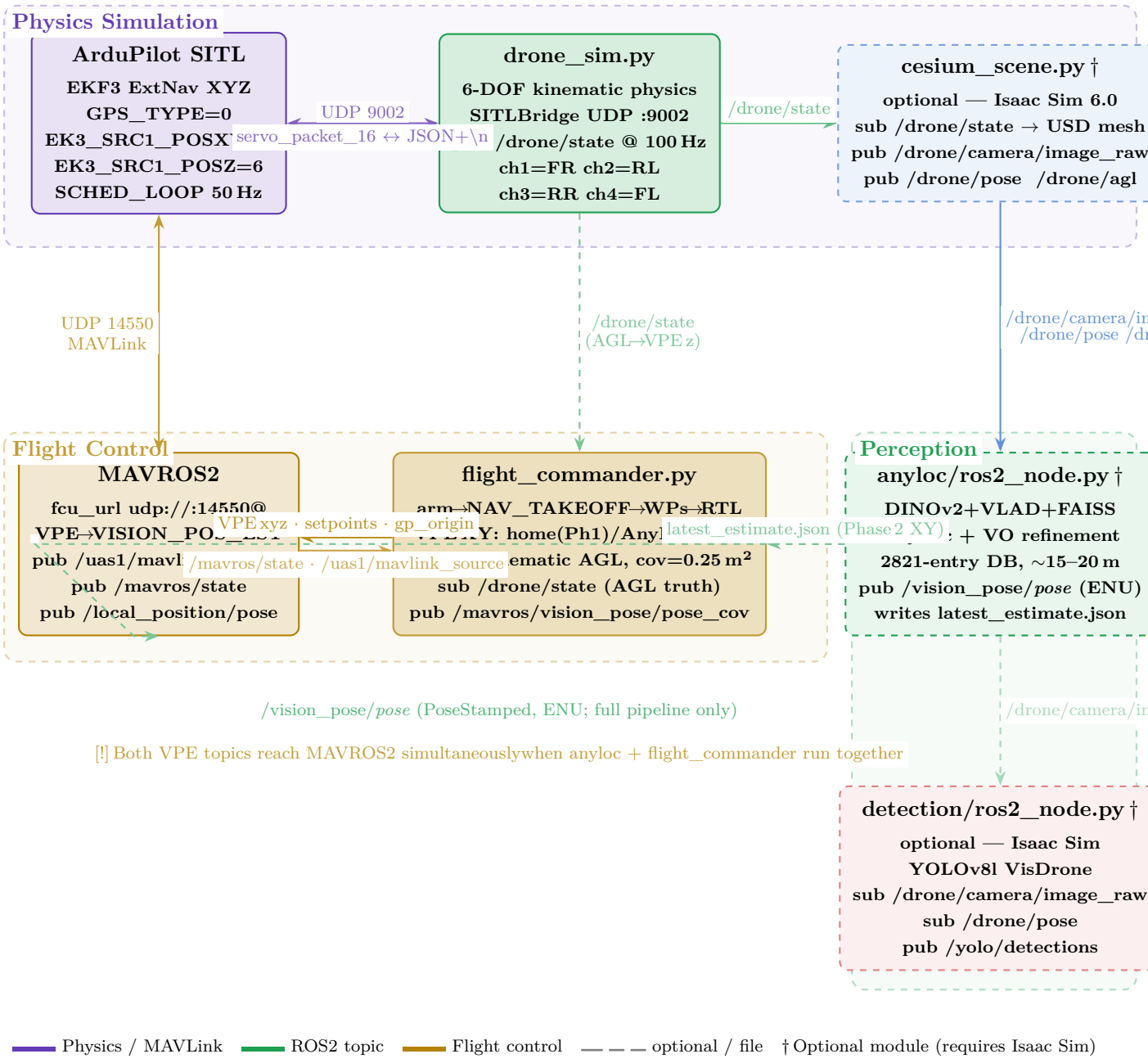


Figure 1: System architecture (updated 2026-06-01, milestone 6j). **Physics simulation layer** (top): `drone_sim.py` runs the 6-DOF kinematic model and exchanges binary servo packets with ArduPilot SITL over UDP 9002; it publishes `/drone/state` (ENU, 100 Hz). Motor channel assignment follows ArduCopter X-frame FRAME_TYPE=1: ch1=Front-Right, ch2=Rear-Left, ch3=Rear-Right, ch4=Front-Left. The optional Isaac Sim visualiser (`cesium_scene.py`) subscribes to `/drone/state` and feeds nadir camera/pose topics to the perception pipeline. **Flight control layer** (middle-left): `flight_commander.py` publishes 3-axis VPE to MAVROS2 — XY from home (Phase 1, below 50 m) or AnyLoc JSON (Phase 2, above 50 m); Z from kinematic AGL read off `/drone/state` (covariance 0.25 m^2). With `EK3_SRC1_POSZ=6` (ExternalNav), ArduPilot EKF3 uses VPE z for altitude — barometer is unreliable in SIM_JSON when `"position"` is omitted. MAVROS2 forwards VPE as `VISION_POSITION_ESTIMATE` and returns status topics. **Perception layer** (right): `anyloc/ros2_node.py` localises from DINOv2+VLAD and writes `latest_estimate.json`; `detection/ros2_node.py` (optional) detects vehicles with YOLOv8l. Dashed borders = optional modules; dashed arrows = optional or file-based interfaces.

Concise Data-Flow Summary

1. **Isaac Sim** renders the Chiayi, Taiwan scene and publishes nadir camera frames (`latest.jpg`) and metadata (`latest_meta.json`) every 5 simulation steps.
2. **SITL Bridge** (`sitl_bridge.py`) receives binary servo packets from ArduPilot on UDP port 9002 and replies with an IMU/velocity physics JSON packet, closing the ArduPilot simulation loop.
3. **AnyLoc** (`run_localizer.py`) runs DINOv2 + VLAD on each new camera frame, matches against a 2,821-entry database (50 m grid), and writes a geo-estimate to `anyloc/latest_estimate.json`. Visual Odometry (Lucas–Kanade) fills in position between full AnyLoc retrievals.
4. **YOLOv8** (`run_detector.py`) detects aerial vehicles in the same frame and returns bounding boxes with class labels.
5. **Vision Bridge** (`run_vision.py`) converts the AnyLoc lat/lon estimate to NED coordinates relative to the SITL home and sends `VISION_POSITION_ESTIMATE` MAVLink messages to ArduPilot EKF3 at 5 Hz.
6. **ArduPilot EKF3** fuses the vision position (configured as *ExternalNav* source, `EK3_SRC1_POSXY=6`) and reports `POS_ABS` in its EKF status flags when the fix is valid.
7. **Orchestrator** (`main.py`, TODO) sends `SET_POSITION_TARGET_LOCAL_NED` commands once EKF3 has a valid position, replacing manual keyboard control.

Module Descriptions

Simulator (`simulator/`)

Status: Working — Cesium terrain, NLSC orthophoto, OSM buildings, quadcopter drone, nadir camera, SITL physics bridge, HUD.

The Isaac Sim scene is centred on **Chiayi, Taiwan** at 23.4509°N, 120.2861°E.

Scene Components

Table 2: Scene data sources

Layer	Source	License
Terrain	Cesium World Terrain (asset 1), quantized-mesh-1.0	©Cesium ion
Buildings	Cesium OSM Buildings (asset 96188), B3DM	©OSM (ODbL)
Imagery	Taiwan NLSC PHOTO2 orthophoto WMTS, zoom 18	©National Land Surveying and Mapping

Drone Model

A quadcopter spanning ≈ 0.8 m is built from USD primitives under `/World/Drone`: *body* (flat cube), *four arms* at 45° intervals, *motor cylinders* at arm tips, *propeller discs* above each motor, and an *orange beacon SphereLight* (5000 cd) for visibility from the overview camera.

Camera and Frame Output

The nadir camera (`/World/Drone/Camera`) uses 18 mm focal length on a 36×27 mm aperture sensor, giving a **90°×73.7° FOV**. At 50 m AGL this covers ≈ 100 × 75 m of ground.

Every 5 simulation steps the renderer writes:

- `drone_frames/latest.jpg` — 640×480 RGB nadir view (ML input)
- `drone_frames/latest_meta.json` — `{step, lat, lon, alt_m, agl_m, centre_elev, yaw_deg, frame_w, frame_h}`

Keyboard Controls

Key	Action
`Tab`	Toggle viewport: overview ↔ drone nadir
`W / S`	Fly north / south (5 m/step)
`A / D`	Fly west / east
`Q / E`	Descend / ascend
`Z / X`	Yaw left / right (1°/step)

Localisation — AnyLoc + Visual Odometry (`anyloc/`)

Status: Working — 2,821-entry database (50 m grid); ≈15–20 m anchor error; geo-constrained search; VO refinement to ≈5–10 m between anchors.

How AnyLoc Works

AnyLoc is a *universal visual place recognition* framework. Given a query image, it extracts patch features with a pre-trained DINOv2 ViT-B/14 backbone, aggregates them into an **intra-normalised VLAD** descriptor ($k = 64$ codewords, $64 \times 768 = 49{,}152$ dimensions), and retrieves the nearest database entry by cosine similarity.

Database Construction

1. Place a 50 m grid (±1500 m from scene centre) → 2,821 positions.
2. For each position, crop the NLSC satellite orthophoto at 50 m AGL (same camera FOV).
3. Extract DINOv2 features → compute VLAD → L2-normalise.
4. Store all 2,821 VLAD vectors in a FAISS `IndexFlatIP` (inner-product = cosine similarity after normalisation).

Geo-Constrained Retrieval

After the first anchor is established, each subsequent AnyLoc retrieval is **constrained to a 200 m geographic window** centred on the current VO-estimated position. Only ≈ 50 of the 2,821 database entries are compared, preventing jumps to visually similar but geographically distant tiles.

$$d_i = \sqrt{(\Delta\text{lat}_i \cdot 111{,}320)^2 + (\Delta\text{lon}_i \cdot 111{,}320 \cdot \cos\phi)^2} \quad \text{filter: } d_i \leq 200\,\text{m}$$

Visual Odometry (VO) Between Anchors

Between AnyLoc retrievals (`ANYLOC_INTERVAL = 10` frames), `VORefiner` tracks Shi-Tomasi corner features using Lucas–Kanade optical flow. Median pixel displacement is converted to Δlat/Δlon via:

$$v_{\text{raw,east}} = -\Delta x_{\text{px}} \cdot m_{\text{px}}^{-1}, \quad v_{\text{raw,north}} = +\Delta y_{\text{px}} \cdot m_{\text{px}}^{-1}$$

$$\begin{pmatrix} \Delta\text{east} \\ \Delta\text{north} \end{pmatrix} = \begin{pmatrix} \cos\psi & \sin\psi \\ -\sin\psi & \cos\psi \end{pmatrix} \begin{pmatrix} v_{\text{raw,east}} \\ v_{\text{raw,north}} \end{pmatrix}$$

where ψ is the drone yaw and m_{px}^{-1} is metres per pixel derived from AGL and FOV.

Accuracy Table

Table 3: Localisation accuracy vs. database grid step

Grid step	Entries	Expected anchor error	Between anchors (VO)
200 m	172	≈65 m	—
100 m	≈688	≈30–40 m	—
50 m (current)	**2,821**	**≈15–20 m**	**≈5–10 m**
25 m	≈11,000	≈8–12 m	—

Hardware used: RTX 2080 Ti; AnyLoc inference: ≈183 ms per anchor frame.

Object Detection — YOLOv8 (`detection/`)

Status: Working — `yolov8l_visdrone.pt` active; auto class-map; fine-tuning pipeline ready.

Active Model

`yolov8l_visdrone.pt` — YOLOv8-Large pre-trained on VisDrone 2019 DET (10 aerial vehicle classes). Confidence threshold: 0.30.

Class Mapping

The `YOLODetector` class builds a class-ID-to-label mapping at load time from `model.names`, making the same code work with both COCO and VisDrone models.

Table 4: VisDrone → canonical class mapping

VisDrone class	ID	Canonical label	Box colour
car	3	car	red
van	4	car	red
truck	5	truck	yellow
tricycle	6	motorcycle	orange
awning-tricycle	7	motorcycle	orange
bus	8	bus	purple
motor	9	motorcycle	orange

Fine-Tuning Pipeline

A complete pipeline adapts YOLOv8 to top-down aerial imagery:

1. `collect_training_data.py` — fly an Isaac Sim grid at 30/60/100 m AGL; export JPEG frames and YOLO labels for 43 synthetic vehicles.
2. `prepare_dataset.py` — download VisDrone 2019 DET (≈7k images); remap to 4 classes; merge with synthetic data.
3. `finetune.py` — 100 epochs with nadir augmentations: `degrees=45`, `scale=0.5`, `mosaic=1.0`, `flipud=0.5`.

Best weights saved to `detection/runs/topdown_v1/weights/best.pt`.

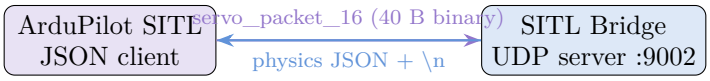
Flight Control — ArduPilot + MAVLink (`control/`)

Milestones 6a–6b-iii: Done — SITL JSON bridge, GPS-denied EKF3, AnyLoc vision position fully fused.

Milestones 6b-iv, 6c, 6d: TODO — autonomous flight commands, IMU reader, IMU anchor validation.

SITL JSON Bridge (`sitl_bridge.py`)

ArduPilot SITL uses a JSON physics backend where ArduPilot is the *client* and the bridge is the *server*:



The physics JSON sent each step:

```
{
  "timestamp": 1234.5,
  "imu": {
    "gyro":        [0.0, 0.0, yaw_rate],
    "accel_body": [sf_bx, sf_by, sf_bz]    # specific force (m/s^2)
  },
  "attitude":  [0.0, 0.0, yaw_rad],
  "velocity":  [vn, ve, vd],               # required=true in SIM_JSON.h
  "rng_1":     agl_m,
  "battery":   {"voltage": 12.6, "current": 5.0}
  # "position" intentionally ABSENT -- AnyLoc provides it via MAVLink
     instead
}
```

Listing 1: Physics state dict sent to ArduPilot

Key design point. The `"position"` field is a GPS substitute in ArduPilot's EKF3. It is deliberately omitted to enforce GPS-denied operation. `"velocity"` (`required=true`) stays to prevent ArduPilot from rejecting the packet.

EKF3 Configuration (`no_gps.parm`)

```
GPS_TYPE        0   # disable GPS sensor
EK3_SRC1_POSXY  6   # horizontal position: ExternalNav (VPE XY)
EK3_SRC1_VELXY  0   # no horizontal velocity source
EK3_SRC1_POSZ   6   # altitude: ExternalNav (VPE z = kinematic AGL)
                    # NOT barometer (1) -- baro stuck at 0 in SIM_JSON
                    # when "position" field is omitted from JSON bridge
EK3_SRC1_YAW    6   # yaw: ExternalNav
VISO_TYPE       1   # enable MAVLink ExternalNav processing
MOT_THST_HOVER  0.5 # hover at PWM 1500 (matches kinematic model)
DISARM_DELAY    0   # enables NAV_TAKEOFF (removes land-detector
   deadlock)
SCHED_LOOP_RATE 50  # 50 Hz loop rate
```

Listing 2: SITL parameter file for GPS-denied operation (no_gps.parm)

MAVLink Controller (`mavlink_ctrl.py`)

`MAVLinkCtrl` is a non-blocking pymavlink wrapper that:

- maintains the latest `HEARTBEAT`, `HIGHRES_IMU`, `ATTITUDE`, `LOCAL_POSITION_NED`, and `EKF_STATUS_REPORT` in memory;
- sends `VISION_POSITION_ESTIMATE` (AnyLoc position → EKF3);
- exposes `arm()`, `takeoff()`, `set_position_ned()` for 6b-iv.

EKF Status Flags

Table 5: EKF_STATUS_REPORT flag bits (ArduPilot EKF3)

Constant	Bit	Hex	Meaning
`EKF_ATTITUDE`	0	0x0001	Attitude valid
`EKF_VEL_HORIZ`	1	0x0002	Horizontal velocity valid
`EKF_POS_HORIZ_REL`	3	0x0008	Relative horizontal position valid
`EKF_POS_HORIZ_ABS`	4	0x0010	Absolute position valid (vision fused) ← *target*
`EKF_PRED_POS_HORIZ_ABS`	9	0x0200	Vision estimate accepted
`EKF_UNINITIALIZED`	10	0x0400	EKF not yet initialised (normal at startup)

Vision Bridge (`run_vision.py`)

Reads `anyloc/latest_estimate.json` and converts to NED relative to SITL home:

$$N = (\phi_{\text{est}} - \phi_{\text{home}}) \cdot 111{,}320, \quad E = (\lambda_{\text{est}} - \lambda_{\text{home}}) \cdot 111{,}320 \cdot \cos\phi_{\text{home}}, \quad D = -(\text{alt} - \text{alt}_{\text{home}})$$

Sends `VISION_POSITION_ESTIMATE` at 5 Hz on TCP port 5763 (separate from the monitor on 5762, since each TCP port accepts one client). Covariance: 20 m horizontal std (≈AnyLoc anchor error), 5 m vertical, 0.3 rad yaw.

Confirmed result: EKF flags reach `ATT,VEL_H,VEL_V,POS_REL,POS_ABS,ALT,PRED_ABS` — full fusion achieved (Milestone 6b-iii).

Stub Bridge (`stub_bridge.py`)

A kinematic hover simulator driven by ArduPilot's PWM outputs. Allows MAVLink testing without Isaac Sim running:

- Integrates thrust from mean motor PWM into vertical velocity and altitude.
- Horizontal position fixed at scene origin (drone hovers in place).
- Runs at 100 Hz; sends the same physics JSON as the real bridge.

Repository Layout

```
no_GPS_drone_project/
+-- instructions/
|   +-- project_plan.md       # module status, design decisions, milestones
|   +-- history.md            # session-by-session change log
+-- simulator/
|   +-- cesium_scene.py       # Isaac Sim scene: terrain + drone + camera + bridge
|   +-- drone_frames/         # live output: latest.jpg + latest_meta.json
|   +-- run_chiayi.sh         # launch script
+-- anyloc/
|   +-- build_database.py     # build VLAD database from NLSC orthophoto (once)
|   +-- localizer.py          # AnyLocLocalizer (DINOv2 + VLAD + FAISS)
|   +-- vo_refiner.py         # VORefiner (LK optical flow)
|   +-- run_localizer.py      # live dual postview + writes latest_estimate.json
|   +-- database/             # 2821-entry VLAD database (49152-dim, 50 m grid)
+-- detection/
|   +-- detector.py           # YOLODetector (auto class-map)
|   +-- run_detector.py       # live annotated postview
|   +-- label_writer.py       # nadir projection for synthetic labels
|   +-- collect_training_data.py  # Isaac Sim synthetic data collector
|   +-- prepare_dataset.py    # VisDrone + remap + merge synth
|   +-- finetune.py           # YOLOv8 top-down fine-tuning
+-- yolov8l_visdrone.pt       # active model (VisDrone 10-class)
+-- control/
|   +-- sitl_bridge.py        # UDP :9002 server --- binary servo in, JSON physics out
|   +-- stub_bridge.py        # kinematic hover stub for MAVLink testing
|   +-- mavlink_ctrl.py       # MAVLinkCtrl + EKF constants + vision + cmds
|   +-- run_mavlink.py        # live terminal EKF monitor (port 5762)
|   +-- run_vision.py         # AnyLoc -> VISION_POSITION_ESTIMATE (port 5763)
|   +-- no_gps.parm           # SITL: GPS_TYPE=0, EK3_SRC1_POSXY=6, VISO_TYPE=1
|   +-- imu_reader.py         # HIGHRES_IMU reader [TODO 6c]
|   +-- imu_fusion.py         # anchor validator + VO gate [TODO 6d]
+-- third_party/
|   +-- ardupilot/            # ArduPilot source; binary: build/sitl/bin/arducopter
+-- main.py                   # top-level orchestrator [TODO]
```

Listing 3: Project directory tree

Running the System

Requirements

- NVIDIA GPU (RTX 2080 Ti or better, 8 GB+ VRAM)
- NVIDIA driver 520+; Ubuntu 22.04 / 24.04
- conda env `isaac_sim_test` (Python 3.12, Isaac Sim 6.0.0)
- Cesium ion account (token embedded in `cesium_scene.py`)
- X11 display (`DISPLAY=:2` or virtual framebuffer)
- System Python packages: `pexpect mavproxy pymavlink future`

Build ArduPilot SITL (one-time)

```
cd third_party/ardupilot
git submodule update --init --depth=1 --recursive   # ~5 min
python3 waf configure --board sitl
python3 waf copter                                   # ~2 min
cd ../..
```

Launch Order (5 terminals)

```
python3 third_party/ardupilot/Tools/autotest/sim_vehicle.py \
    -v ArduCopter --model=JSON --no-rebuild --console --map \
    -l 23.450868,120.286135,46,0 \
    --add-param-file=control/no_gps.parm \
    --out tcp:localhost:5763
```

Listing 4: Terminal 1 — ArduPilot SITL

```
# Full simulation:
cd simulator && ./run_chiayi.sh

# Or lightweight stub (no Isaac Sim needed):
python3 control/stub_bridge.py
```

Listing 5: Terminal 2 — Isaac Sim (or stub)

```
DISPLAY=:2 conda run -n isaac_sim_test python anyloc/run_localizer.py
```

Listing 6: Terminal 3 — AnyLoc localizer

```
python3 control/run_vision.py
```

Listing 7: Terminal 4 — Vision bridge (AnyLoc -> EKF3)

```
python3 control/run_mavlink.py
```

Listing 8: Terminal 5 — MAVLink monitor

Expected EKF Progression

```
TIME      ROLLdeg    PCHdeg     YAWdeg       N m       E m       D m     EKF
    flags
------  ------  ------  ------  ------  ------  ------  ---------
  5.0     0.00     0.00     0.00     0.00     0.00     -5.00  0x0400 UNINIT
 10.0     0.01    -0.02     0.00     0.00     0.00     -5.00  0x0001 ATT
 40.0     0.01    -0.02    90.00     0.12     0.05     -9.87  0x003f ATT,VEL,
    POS_ABS
```

`POS_ABS` (bit 4) confirms that EKF3 has accepted the AnyLoc vision position. Flight commands will now work.

Rebuild the AnyLoc Database

Only needed after the scene or camera FOV changes:

```
conda run -n isaac_sim_test python anyloc/build_database.py --rebuild
```

Milestones

#	Milestone	Status
1	Isaac Sim scene: Cesium terrain + NLSC imagery + OSM buildings	Done
2	Virtual drone + nadir camera publishing frames	Done
3	AnyLoc map database built from simulated views	Done
4	AnyLoc localisation + dual postview on simulated frames	Done
5	YOLO detection working on simulated frames	Done
5a	Switch to VisDrone-trained YOLOv8l; auto class-map in detector	Done
5b	Top-down fine-tuning pipeline (VisDrone + synthetic data)	Ready to run
6a	ArduPilot SITL + JSON bridge (IMU + baro)	Done
6b-i–iv	pymavlink pipeline: GPS-denied EKF3, VPE fusion, flight commands	Done
6e	ROS2 migration: all IPC via topics + MAVROS2	Done
6f	Separate drone physics (`drone_sim.py`) from Isaac Sim	Done
6g	Fix VPE: ENU coordinate order + covariance for EKF POS_ABS	Done
6h	Remove pymavlink; MAVROS2 raw MAVLink; `DISARM_DELAY=0`	Done
6i	NAV_TAKEOFF replaces P-controller; EKF origin blocking; ATC gain reduction	Done
6j	Fix altitude runaway (`EK3_SRC1_POSZ`→6) + 90° course-error (motor layout)	**Done**
6j-wp	Waypoints via position setpoints	WIP
6c	`HIGHRES_IMU` from ArduPilot → localisation pipeline	TODO
6d	IMU fusion: AnyLoc anchor validator + VO quality gate	TODO
7	Full pipeline integrated in simulation	TODO
8	Deploy to real hardware	TODO

Key Design Decisions

ArduPilot before custom IMU fusion. On real hardware, IMU data arrives as MAVLink `HIGHRES_IMU` messages. Building against the MAVLink interface now means *zero code changes* at hardware deployment. ArduPilot's own sensor models (bias drift, noise, temperature) are more realistic than analytical position derivatives.

No numpy — torch tensors throughout. The `isaac_sim_test` conda environment has a dual numpy conflict (pip numpy 2.x vs. conda numpy 1.26.4 from faiss-cpu). All VLAD pipeline operations use torch tensors; numpy is only touched at C-level FAISS call sites where the ABI is compatible.

Geo-constrained AnyLoc search. Unconstrained VLAD retrieval can jump to visually similar but geographically distant tiles (e.g. two similar road intersections 800 m apart). The 200 m window reduces the candidate set from 2,821 to ≈50 entries, making wrong-tile matches geometrically impossible.

Vision position, not bridge position. The JSON bridge's `"position"` field is a GPS substitute in ArduPilot's EKF3. Sending it would give the EKF ground truth, not AnyLoc estimates. It is intentionally absent; AnyLoc position arrives via `VISION_POSITION_ESTIMATE` instead — the same mechanism as Intel RealSense T265 or OptiTrack on real hardware.

Two MAVLink TCP ports. Each TCP port accepts exactly one client. The monitor (`run_mavlink.py`) uses port 5762; the vision bridge (`run_vision.py`) uses port 5763 (opened with `-out tcp:localhost:5763`).

Competition Readiness

The competition (2nd UAV Defense Challenge) requires GPS-denied autonomous UAV operation with multi-target vehicle detection and geo-location.

Table 7: Competition requirement gap analysis

Requirement	Current state	Gap
GPS-denied navigation	AnyLoc + VO (done)	Integrate ArduPilot autonomous flight
Civilian vehicle detection	`yolov8l_visdrone.pt` (done)	Verify car-roof logo detection
Military vehicle detection	Not in VisDrone classes	Collect aerial training data; fine-tune
Oil drums detection	Not in current model	Collect data; fine-tune
Amphibious APC detection	Not in current model	Collect data; fine-tune
Geo-location accuracy (≤50 m)	15–20 m anchor error	Meets requirement (ε_i = 10, full score)
GCS live display	matplotlib windows	Verify HDMI output to judges
Return-to-home	ArduPilot RTL (stub)	Wire up after Milestone 6b-iv

The localisation accuracy of ≈ 15–20 m satisfies the competition's tightest scoring bracket (≤50 m → full 10 position points per target).

Acknowledgements

- **AnyLoc** — "AnyLoc: Towards Universal Visual Place Recognition", IRAL 2024. https://github.com/AnyLoc/AnyLoc

- **YOLOv8** — Ultralytics. https://github.com/ultralytics/ultralytics
- **ArduPilot** — Open-source autopilot. https://ardupilot.org
- **Isaac Sim 6.0** — NVIDIA. https://developer.nvidia.com/isaac-sim
- **Taiwan NLSC PHOTO2** — National Land Surveying and Mapping Center, National Land Surveying and Mapping Center, Taiwan.